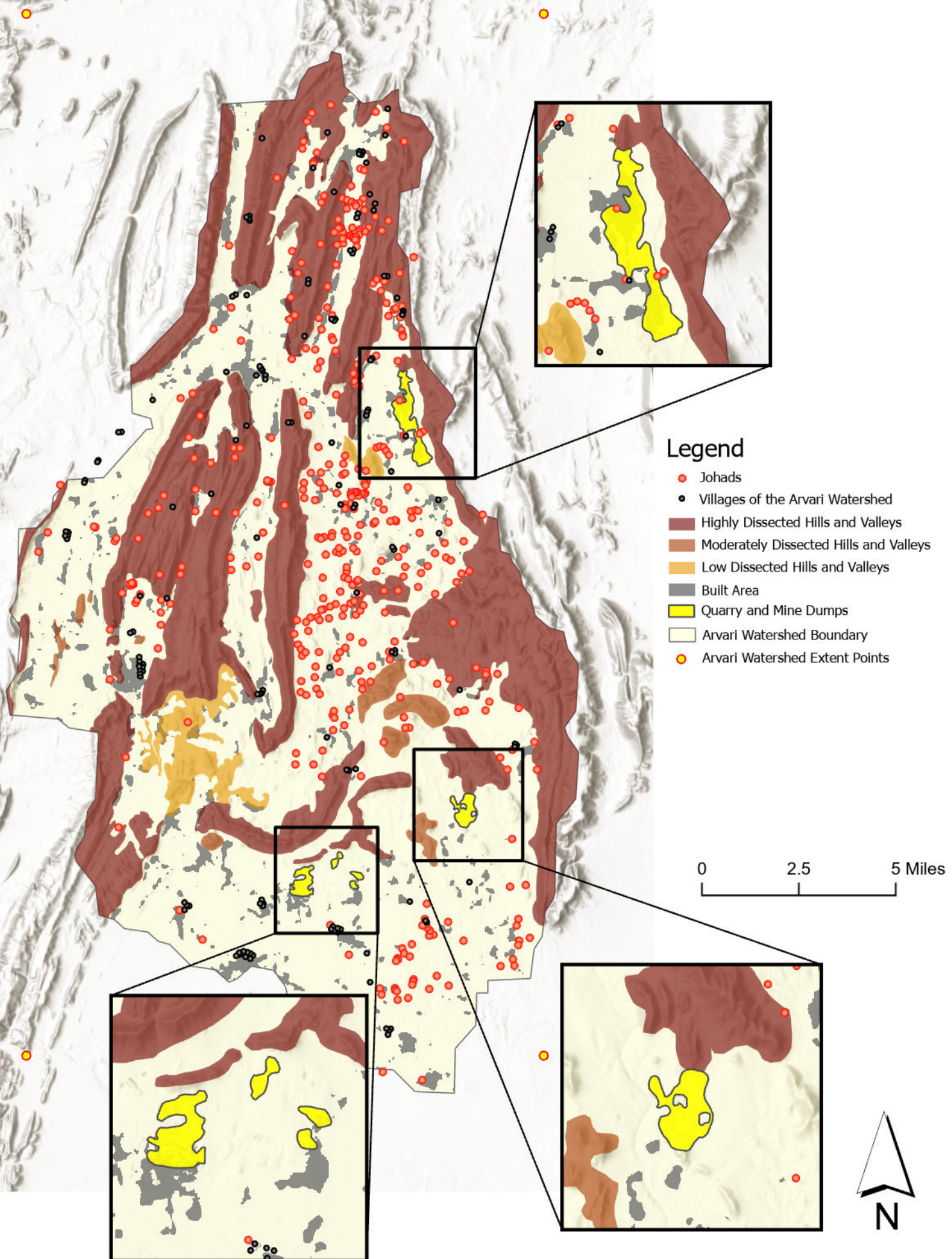


Geomorphology, Quarries, and Mines of the Arvari Watershed



Map #5: Geomorphology, Quarries, and Mines of the Arvari Watershed
Candice Weber, URSP 515

Our collaborators' fieldwork identified the connection of the Arvari River and the Aravalli Mountains to be one of the most important linkages between the ecological processes and socio-economic conditions of the Arvari watershed. Therefore, we have included Map #5 "Geomorphology, Quarries, and Mines of the Arvari Watershed." This map highlights the hills and valleys of the Aravalli range that spread across the entire watershed. The locations of traditional johads and villages are shown on the map along with the location of built-up areas (areas where human settlement is the predominant land cover), quarries, and mine dumps. Inset maps provide a more detailed look at the co-location of quarries (both active and abandoned), johads, and villages in the watershed.

Our collaborators' learned a śloka, or traditional verse, while conducting their fieldwork that describes the relationship between mountains, water, and watershed inhabitants:

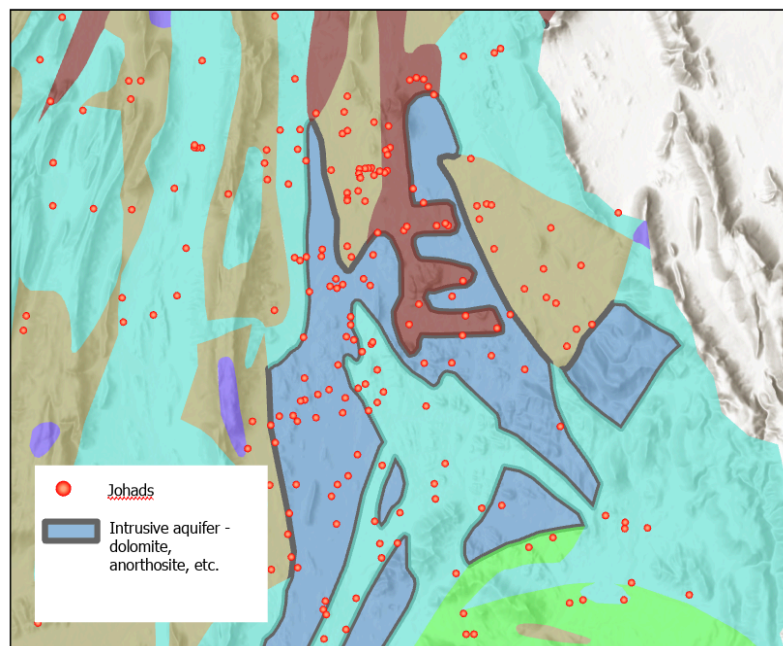
समुद्रवसने देवि पर्वतस्तनमण्डले विष्णुपत्नि नमस्तुभ्यं पादस्पर्श क्षमस्वमे

English Translation:

"O Goddess Earth, one who wears the ocean as her garment and is adorned with mountains as her bosom. O consort of Lord Vishnu, I bow to you. Please forgive me for touching you with my feet."

For the watershed community, the Aravalli Mountains represent the Goddess Earth's breasts giving life through water. The Arvari River itself flows from a source in these mountains to the villages and johads in the valleys below. The Aravalli Mountain range is the oldest mountain range in India. Similar to the Appalachian Mountains in the United States, the Aravalli's were once as high as the present-day Himalayas but over hundreds of millions of years have worn down through erosion. The highest peak in the mountain range today is 1,722 meters.

The mountains shape the hydrogeologic landscape of the watershed. The mountains themselves are underlain with quartzite aquifers while the valleys below have alluvial or intrusive aquifers. Interestingly, a concentration of the watershed's johads are located over and around an intrusive aquifer:



Intrusive igneous rock formations like this aquifer are formed by molten magma intruding into existing rock formations below the Earth's surface and then slowly cooling over time. These intrusions are often in the form of dykes or vertical shafts of igneous rock like dolerite or anorthosite. These dykes then can “act as conduits for groundwater flow within the aquifer. They may also act as semi- to impermeable barriers to the movement of groundwater.”¹

The watershed includes three areas of land marked by both active and closed quarries and mines. These areas are in yellow on Map #5 and highlighted with inset maps. From satellite imagery and also some labels on public maps it is clear these areas are marble and stone quarries, both active and abandoned. Rajasthan produces 85% of the marble in India. These mines can be a threat to human and ecological health. The slurry of water and marble dust produced during cleaning and washing of the stone “reduces soil’s porosity, water absorption and percolation levels. It also affects groundwater recharge. During monsoons, it flows into rivers, drains and other water bodies, affecting water quality, reducing its storage capacity and damaging aquatic life.”²

The Aravalli Mountains are an important biodiversity and climate mitigation resource not only for the Arvari watershed but for all of India. In 2023 the India Ministry of Environment, Forest and Climate Change announced the Aravalli Green Wall Project. This project recognizes the Aravalli Mountains’ role in preventing further land degradation and desertification by acting as a natural barrier between the Thar Desert to the east and watersheds like the Arvari watershed to the west. The project will involve “planting native species of trees and shrubs on scrubland, wasteland and degraded forest land, along with rejuvenating and restoring surface water bodies such as ponds, lakes and streams. The project will also focus on agroforestry and pasture development to enhance the livelihoods of local communities.”^{3,4}

The importance of the Aravalli Mountain range was further demonstrated as recently as December 2025 when the India Supreme Court, under pressure from environmental experts and conservationists in the country, reversed their November 2025 ruling that would have limited the official definition of the Aravalli Mountains to only those hills rising above 100 meters.⁵ Such a rule would have opened the door to even more destructive mining and deforestation in this area. Just prior to the reversal of the ruling, sustainability and environmental management advisor Dinesh Agrawal published a defense of the Aravallis arguing that protecting only the summits of these mountains, and not the hills and valleys below like those of the Arvari watershed, would severely limit the mountains’ role in groundwater recharge, accelerate soil erosion, break wildlife corridors, and “undermine the very objective of preventing desertification.”⁶ The reversal of this ruling will hopefully help preserve the ecological integrity of the Arvari watershed.

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Layers included in Map #5:

Johads and villages layers derived from Tarun Bharat Sangh PDF map digitized by URSP 515 students.

Arvari watershed boundary: Lehner, B., Grill G. (2013): Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. Hydrological Processes, 27(15): 2171–2186. Data is available at www.hydrosheds.org.

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Other layers reviewed but not included in Map #5:

Esri India Content. "Aquifer Systems of India" [Feature layer] Scale not given. "The Aquifer Systems dataset, compiled by the Central Ground Water Board (CGWB) of India, developed under the framework of the National Project on Aquifer Management (NAQUIM)." September 25, 2025. <https://vcu.maps.arcgis.com/home/item.html?id=fb1d1156abeb4334a6608d045f0ee484>. (April 10, 2026).